A REVIEW OF BIOMARKERS OF ENVIRONMENTAL ENTERIC DYSFUNCTION IN LOW- AND MIDDLE-INCOME COUNTRIES

Muo Gabriel Paul [1,2] Evans Mudibo[1] James Njunge[1]

[1] KEMRI-Wellcome Trust Research Programme, Kilifi [2]Department of Health and Human Sciences, Pwani University

INTRODUCTION

- Environmental Enteric Dysfunction (EED) is a subclinical chronic disease of the small intestines widespread among children in Low- and Middle-Income Countries (LMICs).
- Characterised by inflammation, malabsorption, and gut permeability, EED is associated with stunting and slowed cognitive development.
- Histologic features include villus blunting, crypt hyperplasia, intraepithelial lymphocytosis, infiltration of the Brunner's glands, and Paneth Cell depletion.
- Diagnosis is challenged by a lack of standardized non-invasive biomarkers.

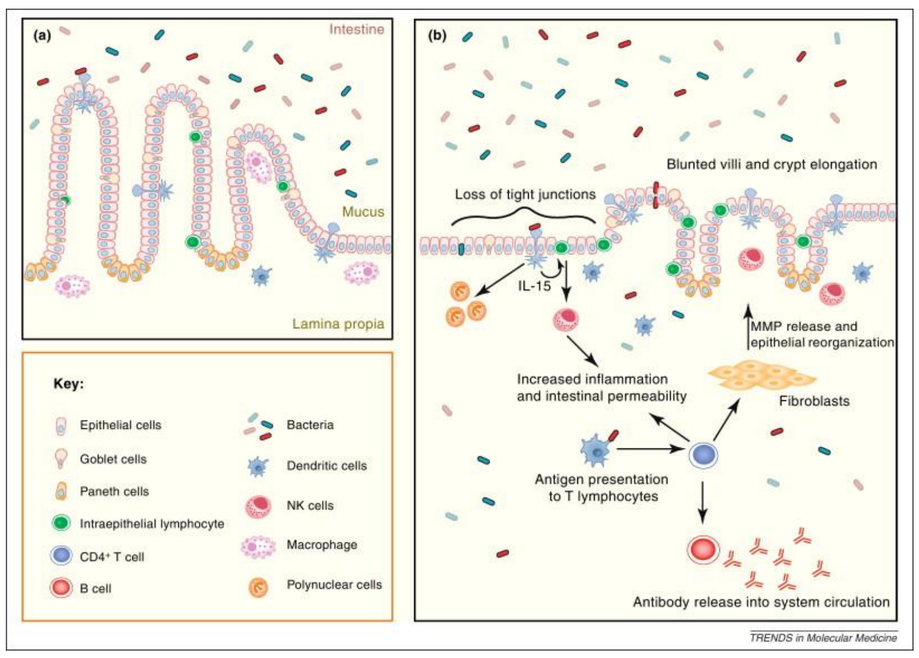

Figure 1: The proposed pathogenesis associated with environmental enteropathy (Korpe et al.,2012) a. Normal gut b. Deranged gut

RESEARCH QUESTION

- What is the distribution pattern of biomarker concentrations among children under 5 years in LMICs?
- Are there optimal cutoffs established in LMICs for use among children under 5 years?
- What proportion of children and/or studies have elevated biomarker levels based on cutoffs?

METHODOLOGY

Narrative Review

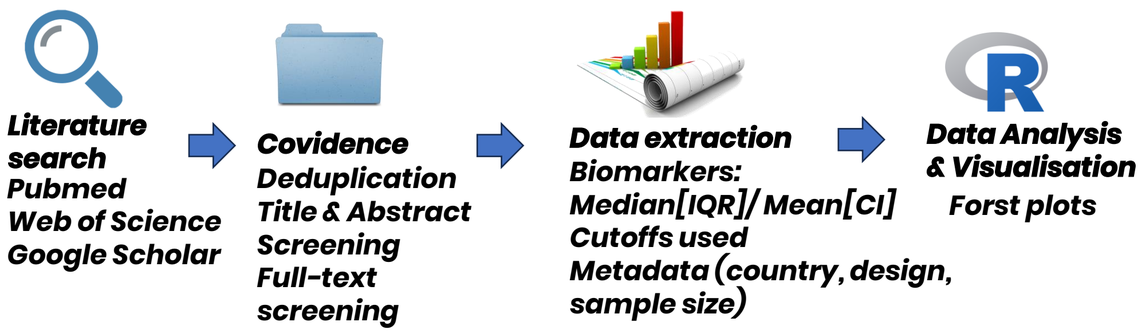

Figure 2: Illustration of literature search, screening and selection of studies

RESULTS

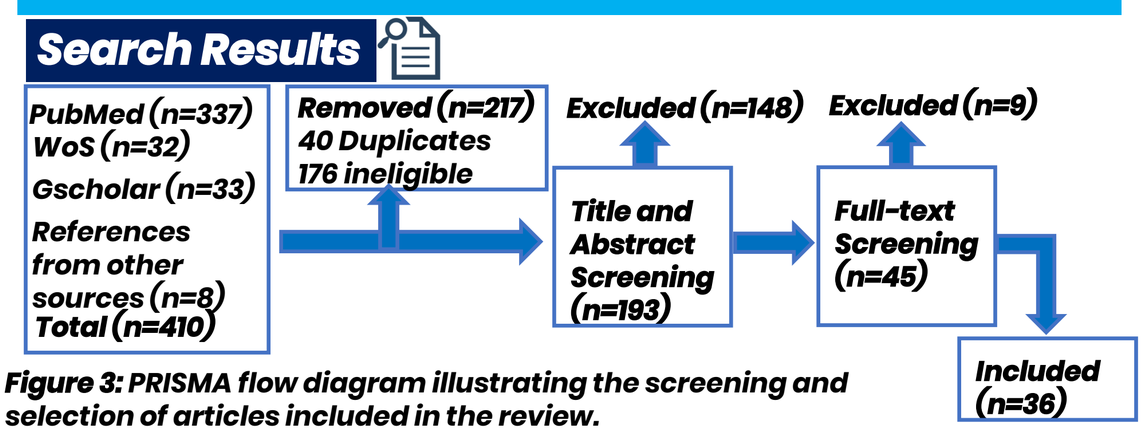

Figure 3: PRISMA flow diagram illustrating the screening and selection of articles included in the review.

- 36 Studies
- Done in 22 LMICs
- Involving 17,430 children under 5

Figure 4: Map of countries (shaded dark-grey) included in this review

Table 1: Findings on cutoffs used in the LMICs and proportions of studies with elevated cutoffs

Biomarker	Cutoff Utilized	Origin of Cutoff	% of Studies Exceeding Cutoff	Key Contextual Note on Cutoff
NEOPTERIN Intestinal Inflammation (Th1 cell activity)	<70 nmol/L	High-Income Country (HIC) setting (Austrian adult controls/colon dialysate study)	100% of the 15 studies reported median concentrations above this threshold.	Cutoff would trigger investigations for inflammatory bowel disease (IBD) in HICs. All studies utilizing this cutoff reported high concentrations.
ALPHA-1 ANTITRYPSIN Intestinal Permeability/Protein Loss	<270 µg/g	HIC setting (Derived from children in the USA)	84% of the 19 studies reporting median concentrations exceeded this threshold.	None of the studies established AAT cutoffs within LMIC settings.
LACTULOSE TO MANNITOL RATIO Intestinal Permeability and Absorption	<0.07	Both HIC and LMIC settings (First defined among Guatemalan infants)	83% of the studies reporting median concentrations were above this threshold.	This cutoff, though LMIC-derived, showed high rates of elevation.
MYELOPEROXIDASE Intestinal Inflammation (Neutrophil activity)	<2000 ng/mL	HIC setting (Derived from Japanese IBD cases)	81% of the 21 studies reporting median concentrations exceeded this threshold.	No study has attempted to generate MPO cutoffs in the LMIC setting.
CALPROTECTIN Intestinal Inflammation (Neutrophil activity)	Varied; frequently <150 µg/g (or <50 µg/g)	Varied (Age-specific LMIC cutoffs exist, but HIC thresholds for IBD are commonly used)	71% of the 14 studies reported median concentrations exceeding the 150 µg/g threshold.	One study generated age-specific cutoffs in Ugandan children (e.g., 249 mg/kg for 0 < 1-year-olds).

Sample Forest Plots on Distribution of biomarker concentrations

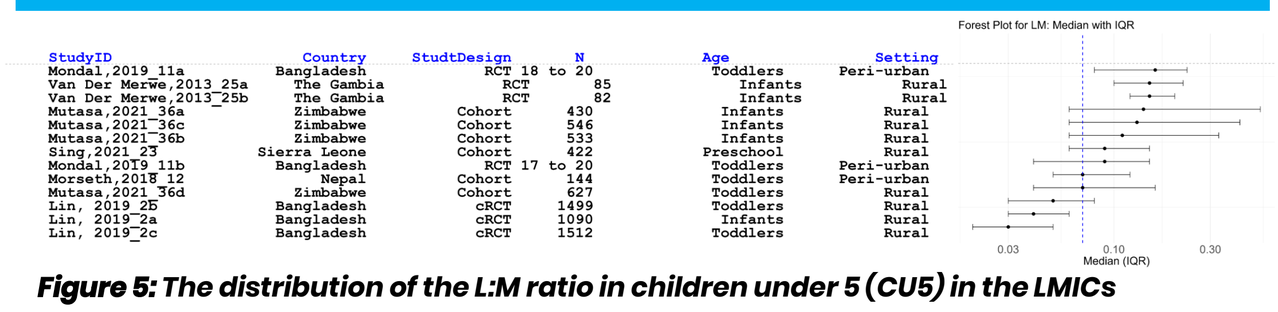

Figure 5: The distribution of the L:M ratio in children under 5 (CU5) in the LMICs

CONCLUSION

The review highlights generally high levels of EED biomarkers in children under five in LMICs, based on HIC standards. Cutoffs adopted would trigger inflammatory bowel disease investigations in HICs. Establishing relevant LMIC-specific cutoffs is crucial, as EED interventions could greatly impact growth and development.

ACKNOWLEDGEMENTS

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